

JANAV RANA

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Codeforces: JanavRana

Education

Bachelor of Technology (B.Tech.) in Computer Science and Engineering
Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
CGPA: 7.89

Technical Skills

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Organization and Architecture

Programming Languages: C++, Python, C, JavaScript, SQL, PL/SQL

Web Development: HTML, CSS, JavaScript **Computer Vision:** OpenCV, MediaPipe

Libraries: NumPy, Pandas, Matplotlib **Tools:** Git, GitHub, VS Code

Projects

Visual Engagement Analysis System

Technologies: Python, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib

- Built a real-time computer vision system that estimates visual engagement using facial landmark analysis.
- Engineered a modular pipeline for blink detection, gaze estimation, head pose estimation, attention scoring, session logging, and HTML report generation.
- Implemented personalized calibration and a rule-based attention scoring engine to improve robustness across different users.
- Generated detailed session analytics, including attention trends, blink statistics, gaze patterns, and interactive HTML reports while maintaining real-time performance on CPU-only systems.

Genetic Algorithm for EV Charging Station Optimization

Technologies: Python, Genetic Algorithm

- Developed an optimization model for efficient EV charging station placement.
 - Implemented genetic algorithm operations including population initialization, fitness evaluation, selection, crossover, and mutation.
 - Performed simulations to evaluate algorithm performance under different parameter configurations.
 - Applied optimization techniques to address a real-world infrastructure planning problem.
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Relevant Coursework

- Actively pursuing self-driven learning in software development, algorithms, and artificial intelligence through hands-on projects.
- Developing strong analytical and problem-solving skills by solving algorithmic challenges and optimizing solutions.